

Photosynthesis and Plant Biology

Photosynthesis is the process by which plants convert light energy into chemical energy stored in glucose.

Chlorophyll is the green pigment found in chloroplasts that absorbs sunlight for photosynthesis.

Stomata are small pores on the surface of leaves that regulate the exchange of gases like carbon dioxide and oxygen.

The Calvin cycle uses carbon dioxide and energy from ATP to produce glucose within the chloroplast stroma.

Mitochondria are organelles responsible for cellular respiration, converting glucose into usable energy.

Transpiration is the process by which water moves through a plant and evaporates from leaves, stems, and flowers.

Xylem tissue transports water and dissolved minerals from the roots upward to the rest of the plant.

Phloem tissue transports the sugars produced during photosynthesis to other parts of the plant for growth and storage.